Find the Promos: A Tool That Reads a Microdrama and Picks Its Best Ads

Kuku AI Generalist Assignment — Problem 1

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Prototype (code + run steps): `github.com/krishagarwal-22/microdrama-promo-finder`

Raw outputs (all 50 episodes): `.../microdrama-promo-finder/tree/main/out`

Why this problem, and who it serves

Today an editor hand-picks promo moments from every episode of every show, one at a time, and new shows ship weekly. The work grows linearly with the catalogue, making it the clearest bottleneck of the three briefs, and it has the richest signal to work with: timestamped screenplays I can read deterministically. The audience is also well-defined — US microdramas are driven by women aged roughly 30–60, documented as especially drawn to romance and revenge, and this show is exactly that: a betrayal-at-the-altar revenge drama. That told me not only *where* to cut, but *which* moments this audience rewards. The tool proposes a ranked shortlist of promo clips with exact in/out timestamps, turning the editor's job from "watch everything and find" into "review a few and approve."

What it does

Point it at the scripts and it returns, per promo: the episode, the start/end timestamp of each segment, the hook (first three seconds), the cliffhanger it cuts on, one line on why it works, and a 0–100 score; then it ranks all promos across the show. On the pack it found 97 promos across all 50 episodes in under a second, 28 of them carrying a full wrong-then-payback arc.

How it works

Free, deterministic rules do the finding; the judgment step is isolated so a model can be swapped in. It parses the scripts (regular, with a type, speaker, emotion tag and in/out time per beat — no AI needed), scores each beat as a potential hook (high-arousal emotion, shock words, short punchy lines) and cliffhanger (unanswered question, cut to black, score swell, episode end), and for each strong hook takes the best cliffhanger 20–60s later as a candidate. Candidates are judged on a craft rubric (hook 35%, cliffhanger 30%, stands-alone 20%, intensity 15%) and then an audience-resonance layer that lifts injustice-to-vindication clips. Finally it drops overlaps, keeps the best one or two per episode, and ranks the rest. The flag `--scorer llm` swaps the judge for an API model.

Why these numbers are grounded

- **3-second hook:** platforms treat the first 3s as a distinct ranking signal; $\sim$40% of viewers swipe within it and $>$70% decide whether to stay (TikTok retention data).
- **20–60s length:** the platform sweet spot; $\sim$72% of TikToks run under 60s and reach falls off past 90s.
- **Cliffhanger:** Loewenstein's information-gap theory (1994) and the Zeigarnik effect (1927). Honest caveat: Zeigarnik's recall claim replicates weakly, so I lean on tension, not memory.
- **Which emotions:** Berger & Milkman (2012, *JMR*) — high-arousal emotions travel; low-arousal sadness does not.
- **Audience resonance:** Affective Disposition Theory (Weber et al., 2008, tested on 500+ female serial-drama viewers) — enjoyment peaks on poetic justice: the wronged, liked character vindicated, the villain punished.

Approach and scale

The expensive judgment — watching — is replaced by cheap rules that do the finding; a model is optional and only re-ranks a short list. The core has no dependencies and runs instantly, so cost is dominated by the optional model call. At real volume (20 shows/week $\times$ 50 episodes, $\sim$2 candidates each) that is $\sim$2,000 short re-rank calls a week on a mini model — cents, not an editor-week. The model is a one-flag swap via `PROMO_LLM_MODEL` and falls back to the free scorer if no key is set, so a batch never breaks and a stronger model drops straight in. A human stays in the loop: the tool proposes ranked clips with exact timestamps; the editor approves or nudges the in/out points.

How I know it is good, and where it fails

I ran it on the full pack and read the top picks against the scripts. #1 (Ep 38) opens on "Do you even know what happened to me?" with a full wrong-then-payback arc; #2 (Ep 18, 02:38–03:12) opens after the arson on "Someone wanted us out." and cuts to black on a vow of vengeance. The poetic-justice layer re-ranked 11 of the top 15 toward betrayal-and-vindication clips while keeping the verified-strong ones. Across the pack: durations 20–59s (median 38), every episode covered, 36/97 cut on an unanswered question, 28/97 carry a full arc. Weaknesses:

- The resonance signal is keyword-based and can miss justice that is *shown, not said*; an aggressive weight once buried the strong Ep 18 clip, so I dialled it to 0.35. This is the clearest case for the swappable LLM scorer.
- "Stands-alone" is coarse (only 0.70 or 0.85); a model judges it far better.
- Purely visual hooks (a silent slap) score low; intensity is a weak differentiator, kept at low weight and flagged.

Tools used, and one place AI got it wrong

I used Claude to scaffold the parser/engine, pressure-test the scoring, and gather the audience and psychology sources, which I then checked. One concrete miss: its first intensity metric scored almost every clip at $\sim$0.10 — too harsh, and the scripts tag emotion mainly on dialogue — so intensity was near-useless. Rather than fake a signal I cut its weight and documented it. The scoring design, the continuous 20–60s clip decision, grounding the weights in the audience and Affective Disposition Theory, and the hand-tuning of the resonance weight are mine.

Run: `python find_promos.py "<scripts_dir>" --out out` $\rightarrow$ `promos_ranked.md`, `promos.json`, `per_episode/`. Full citations in `PSYCH_NOTES.md`. The input pack is Kuku's and is neither committed nor hosted; the demo app takes uploads at runtime.